

3D Printing, Limb Prosthetics and Generative Design: A Scoping Review

Kai Xu, Sheng-Feng Qin
Arts, Design and Social Sciences
Northumbria University
Newcastle Upon Tyne, United Kingdom

k.xu@northumbria.ac.uk; sheng-feng.qin@northumbria.ac.uk

Abstract— This review has investigated the gap in the literature on utilizing Generative Design tools in the process of 3D Printing (3DP) based Limb Prosthetics (LP) fabrication. From this review, we identify there is great potential to address the leading challenges associated with 3D Printed Limb Prosthetics: Cost and Weight. This emerging technology also has the potential to allow deeper customization of the limb prosthetic to be performed by a technician without professional 3D Modelling training.

Keywords—3D Printing; Limb Prosthetics; Generative Design; Topology Optimization; Parametric Design

I. INTRODUCTION

Research into 3D printed (3DP) limb prosthetics (LP) has gained pace in recent years since the rapid development of Parametric design (PD) and Generative Design (GD), which allows high-quality parts to be fabricated at a lower cost (both time and price) by a technician with minimal 3D modelling training [1].

Limb prosthetic (LP), since its earliest development in 1500 BC, has been the standard approach towards enabling individuals who have lost a limb to regain their independence [2,3] and is widely recognised as significantly improving the quality of life (QOL) of amputees after the rehabilitation process [4]. However, there are significant amounts of complex interdisciplinary challenges involved in the process of using LP for limb loss rehabilitation [5], one of the most widely discussed issues is the cost of this procedure [6] and the discomfort of using LP[6].

Building on recent research that has discussed the great potential of 3DP to address the price and comfort issues associated with the use of prosthetics for limb loss rehabilitation, this study will continue to fill the research gap in this area by introducing Generative Design methods. These methods could make the design process involved in this practice easier thus allowing the practice to be more widely available. These methods could also lead to higher-quality designs which could reduce the need for high-end printers and improve the comfort of printed parts.

One of the main aims of this research is to investigate whether Generative Design (GD) and 3D Printing (3DP) can provide a better experience for the prosthetic user while reducing the manufacturing and maintenance costs of the upper or lower limb prosthetic.

This chapter will outline the most recent developments, key discoveries and current state of the art of Generative Design (GD), 3D Printing (3DP) and Limb Prosthetics.

II. METHODOLOGY

Utilising the methodological framework for conducting scoping studies proposed by Arksey and O'Malley [7] and the improved five stages of the review process proposed by Levac et al. [8], we conduct this review to investigate the current status of generative design research in the field of three printed prostheses. We conducted this research built on the scoping review conducted by Ribeiro et al. [3] and the systemic review conducted by Diment et al. [9].

Ribeiro et al.[3]'s work has identified that the subject of 3D printing and amputation still requires more trials and research to generate concise unbiased views on whether 3D printed prosthetic devices are more comfortable for patients, more functional and more attractive to look at, and Ribeiro et al. [3] has also confirmed this with their systematic review, stating that despite the optimism and enthusiasm in the field of rehabilitation medicine for the 3DP prosthesis, it is recommended that we should still conduct randomised clinical trials to assess the efficacy and effectiveness of the 3DP prosthesis before incorporating the 3DP upper limb device into standard clinical practice. This research will build on their studies to identify the gap in the literature when Generative Design (GD) is added into consideration.

Scoping reviews are different from literature reviews,. Scoping reviews are particularly suited to disciplines with emerging evidence such as rehabilitation science. In particular, the lack of randomised controlled trials makes it difficult for researchers to conduct systematic reviews, scoping studies can incorporate a range of study designs in the published and grey literature to create a structure that can be used as a reference framework for future systematic studies.

Although Ribeiro et al. [3]'s work has already identified the importance and necessity to continue evaluating the performance, improvements and significance of 3D printed parts in comparison to traditional casting methods, this study argues that on this basis the implications and improvements of Generative Design (GD) in 3D printed (3DP) prosthetic parts also need to be reviewed and evaluated for their significance and improvements.

A. Stage 1 Identifying the research question (RQ)

The core research question of this review was to investigate the state of knowledge regarding utilising Generative Design (GD) for 3D Printed (3DP) Limb Prosthetics (LP). The objectives of this scoping review were to investigate (RQ1) how much research has been conducted in this area; (RQ2) what generative design methods they have

3DP	((‘three dimensional’ OR ‘Three-dimensional’ OR ‘3 Dimensional’ OR 3D OR ‘3-D’) AND (printing OR print OR printed)) OR ‘3D-printing’ OR ‘3D-print’ OR ‘3D-printed’ OR stereolithography OR ‘laser melting’ OR ‘electron beam melting’ OR ‘fused deposition modelling’ OR ‘fused deposition modelling’ OR ‘selective laser sintering’ OR ‘additive manufacturing’ OR ‘rapid manufacturing’ OR ‘rapid prototyping’ OR ‘layered manufacturing’ OR ‘digital manufacturing’ OR ‘3D prototyping’ OR ‘3D fabrication’ OR ‘rapid fabrication’ OR ‘freeform fabrication’
LP	‘prostheses and implants’ OR prosthetic OR prosthetics OR prosthesis OR prostheses OR ((artificial OR bionic OR robotic) AND (arm OR finger OR hand OR wrist OR leg OR foot OR foot)) OR transradial or ‘trans-radial’ or transhumeral or ‘trans-humeral’ OR ‘upper- limb’ OR ‘upper limb’ OR ‘upper-extremity’ OR ‘upper extremity’ OR ‘lower- limb’ OR ‘lower limb’
GD	(‘Generative Design’ OR ‘Topology optimization’ OR ‘Topology optimisation’ OR ‘Parametric Design’)

Figure 1. Step 1 Search Criteria

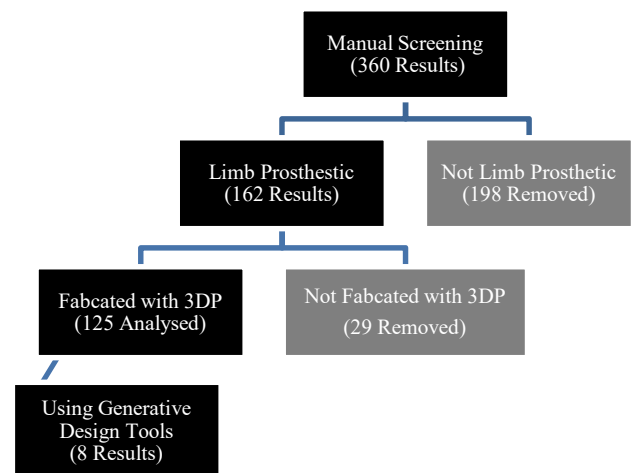


Figure 2. Manual Screening Process

used; and (RQ3) how the use of generative design may affect the outcome of 3D printed limb prosthetics.

B. Stage 2 Identifying relevant studies

This study conducted a search of all articles available in the Web of Science Core Collection up to 28 April 2022 and was aided by searches using Google Scholar. To ensure that the results were comprehensive, the search was divided into two steps.

First, this research retrieved all literature related to both '3D Printing (3DP)' and 'Limb Prostheses (LP)'. We then proceeded to manually screen the literature related to both '3D Printing (3DP)', 'Limb Prostheses (LP)' and 'Generative Design (GD)'. See Figure 1 below for the search criteria that have been used in Step 1.

After removing duplicates, we found 541 papers that met both of the first two parameters (3DP AND LP) in the title field; 4640 papers that met both of the first two parameters (3DP AND LP) in the topic field; 1 paper that met all of the three parameters (3DP AND LP AND GD) in the title field and 100 papers that met all of the three parameters (3DP AND LP AND GD) in the topic field. As there were too few search results when the GD criteria were added, we decided to perform a manual filter from the results that satisfied both 3DP and LP.

C. Stage 3 Study selection

From the 541 results we then excluded all papers related to 'Auricular'; 'Avenue For Resident Education'; 'Chest-Wall Bony Defects'; 'Complete-arch'; 'Cranioplastic'; 'Crowns and Fixed Prostheses'; 'Dental'; 'Dentistry'; 'Drilling Guides'; 'Edentulous Maxilla'; 'Facial'; 'Femoral Shaft Prostheses'; 'Full-arch Immediate Loading'; 'Glenoid Destruction With Cuff Arthropathy'; 'Hip Revision Surgery'; 'Jaw'; 'Joint Replacement Surgery'; 'Limb Salvage Surgery'; 'Limb Salvage Surgery'; 'Maxillectomy'; 'Maxillofacial'; 'Microtia Reconstruction'; 'Modular Hemipelvic Prosthesis'; 'Musculoskeletal Disorders'; 'Nasal'; 'Obturator'; 'Oligodontia'; 'Oral'; 'Orbital prosthesis'; 'Orofacial'; 'Patient-Specific Femoral Prosthesis'; 'Polyether-Ether-Ketone

(PEEK) Prosthesis'; 'Pubic Bone Defect'; 'Reconstruct A Cervical Lateral Mass'; 'Segmental Mandibular Reconstruction'; 'Single-Sided Lateral Wedge Insole'; 'Skull'; 'Stapes Prosthesis'; 'Superior Canal Dehiscence'; 'Swing-lock Prosthesis'; 'Temporomandibular'; 'Total Hip Arthroplasty'; 'Total Hip Arthroplasty'; 'Tracheoesophageal' or 'Unicondylar Femoral Defects', 360 results remain.

These results were not excluded from the first search due to the fact that terms such as 'prosthesis', and 'robotic arm' have vague but broad interdisciplinary definitions and their exclusion at the initial stage will result in the exclusion of many important LP-related literature. Therefore, a manual screening process was performed after (see Figure 2).

By analysing the abstracts of 360 remaining papers, 198 papers that were not LP were excluded from the results, after this step, 29 studies that did not use 3D printing as a manufacturing method were excluded from the results. We analysed the remaining 125 studies in full-text and additionally flagged 8 studies that used generative design as a design tool.

D. Stage 4 Charting the data

This study next manually extracted basic information about these studies; the 3D printing methods they used; the types of prostheses involved; the research questions; and the main findings are collated into a database to allow for the following analysis. (See Table 1 Summary of included articles)

E. Stage 5 Collating, summarizing, and reporting the results

After analysing the selected literature in full, we have grouped them into two areas of discussion: the first is using generative design tools to improve the performance of 3D printed prostheses, such as weight reduction or improving robustness; the second is using generative design tools to allow the design of prostheses to be changed by technicians without specific training in 3D modelling.

Table 1. Summary of included articles

Authors	Year	3DP Technology	3DP Material	Design tools	Simulation tools	Initial 3D model source	Shape optimization method	3DP post-processing	Amputation Type
Lai, C. W.; Hsu, L. H.; Huang, G. F.; Liu, S. H.	2009	FDM	Undeclared	C++, Open Graphic Library, CATIA	Undeclared	Laser scan 3D modeling	FE Analysis, Parametric design	Reinforced with a layer of resin	Prosthetic socket (transibial)
Objective	Develop an interface system that allows the prosthetician to modify the shape of the stump model to achieve an acceptable pressure distribution in the pressure tolerant (PT) and pressure relief (PR) regions. Improves the complex process and inconsistent quality of traditional methods of making prosthetic sockets.			Outcome	This study develops an interface system that allows the prosthetist to modify the shape of the stump model in order to achieve acceptable pressure distribution in the PT and PR areas. It reduces the training time of the prosthetist, makes it easier to modify the pressure tolerance and retention zones, it can store a digital model of the socket and allows new sockets to be made according to the patient's requirements. This system has shown that a prosthetic socket can be designed using scanned points from the residual limb. An initial socket can then be made from this socket model using the 3DP machine and then strengthened by wrapping it with a layer of resin.				
Peng, H. K.; Hsu, L. H.; Huang, G. F.; Hong, D. Y.	2009	FDM	Undeclared	CATIA	MSC Marc	Laser scan 3D modeling, CAD modelling	FEM simulation	Reinforced with a layer of resin	Prosthetic socket (transibial)
Objective	Validating the suitability of a newly developed resin-reinforced 3D printed prosthetic socket.			Outcome	This study was able to design the geometry of a prosthetic socket by using FE analysis to assess the interface pressure between a resin-reinforced RP socket and the stump. The results of the interfacial pressure measurements indicate that the FE simulation model is effective in supporting socket design.				
Mutlu, Rahim; Alici, Gursel; Panhuis, Marc In Het, Spinks, Geoffrey M.	2016	FDM(UP Plus 2)	TPU	Undeclared	ANSYS	CAD modelling	FEM simulation	Undeclared	prosthetic finger
Objective	Designing a soft, monolithic prosthetic finger that can be printed at low cost.			Outcome	A soft, monolithic prosthetic finger that can be printed at low cost was developed through finite element analysis and fused deposition modelling. Experiments and finite element analysis confirmed that an asymmetric elliptical bending hinge is the most suitable type for use in a soft prosthetic robotic finger. After estimating the effective modulus of elasticity, the bending of a soft, monolithic robotic finger was simulated, which showed good correlation with actual experimental results.				
Tao, Zhen; Ahn, Hyoung-Jong; Lian, Chenglong; Lee, Kwang-Hee; Lee, Chul-Hee	2017	FDM	PLA	Siemens NX	HyperMesh, OptiStruct 13.0	CAD modelling	Topology optimization, FE Analysis	Undeclared	prosthetic foot
Objective	Developing a passive prosthetic foot that is designed and optimised by topological optimisation, finite element analysis and experiments on its 3D printed model to validate the structure.			Outcome	A passive prosthetic foot has been designed and optimised through topology optimisation. The structure was validated by finite element analysis and experiments to simulate heel strike (HS) and toe off (TO) boundary conditions, resulting in a 62% weight reduction of the proposed model compared to the initial model.				

Table 1. Summary of included articles

Authors	Year	3DP Technology	3DP Material	Design tools	Simulation tools	Initial 3D model source	Shape optimization method	3DP post-processing	Amputation Type
Bustamante, Marlene; Vega-Centeno, Rodrigo; Sanchez, Midori; Mio, Renato	2018	FDM	ABS	Autodesk Inventor 2016, Rhinoceros 5.0, MS Excel	Undeclared	CAD modelling	Parametric design	Undeclared	prosthetic hand
Objective	Developing a prosthetic hand design that can be changed by changing the parametric dimensions of the finger phalanges and palms on an Excel sheet.Using connecting rods instead of cables as finger flexors and a new cable adjustment system to make the stretching of finger tendons easier and faster.			Outcome	A prototype of a new body-dynamic parametric hand prosthesis has been developed that allows the original design to be customised for different hand sizes by varying the dimensions of each part. The resulting model is aesthetically pleasing and practical, and the small wrist flexion angle required for the hand to activate the fingers makes it easy to use and comfortable.				
Lim, Daniel; Georgiou, Thomas; Bhardwaj, Aashish; O'Connell, Grace D.; Agogino, Alice M.	2018	FDM(Ultimkaer 2.0+)	PLA	Autodesk Fusion 360	MATLAB	CAD modelling	Parametric design	Undeclared	prosthetic finger
Objective	Investigate the relationship between the width and length of the index finger, to propose a parametric model to better match the dimensions and range of motion of the original finger, and to produce a model based on the measurements to verify the feasibility and suitability of using additive manufacturing.			Outcome	A correlation between the width and length of the finger was derived from measurements of 50 participants, by implementing parametric modelling that controlled both parameters (width and length) independently. A crossed four-link mechanism was used to produce a single degree of freedom, with the ROM equation for the effective fingertip derived from the Freudenstein equation. The ROM determined by parametric modelling was shown to optimise the ROM path by removing dependencies between parameters. The feasibility of manufacturing parts by 3D printing techniques with parameters designed by automated CAD and the integrity of the model assembly were verified.				
Leonardis, Daniele; Frisoli, Antonio	2020	FDM (Makerbot 2X)	TPU 95A filament(by Ultimaker), Polyethylene Terephthalate Glycole(PETG)	Undeclared	Undeclared	CAD modelling	FEM simulation	Reinforc printed parts with metallic materials	prosthetic hand
Objective	Developing a robotic hand designed for ease of manufacture and maintenance, robustness and compliance in a practical operating environment.			Outcome	A robotic hand that meets specific design guidelines is designed and initially empirically validated. It can be easily produced and maintained at low cost and, despite being limited by a small number of degrees of freedom, it can still successfully grasp common objects of different sizes and handles in experiments. Its robustness in overcoming lateral collisions was verified in terms of its ability to recover from apparent deformations and loose tendon transfers.				
Vijayan, V.; Kumar, S. Arun; Gautham, S.; Masthan, M. Mohamed; Pirachudan, N.	2021	FDM	PLA	Undeclared	ANSYS	CAD modelling	Topology optimization	Undeclared	prosthetic foot
Objective	Developing a lightweight, robust 3D printed prosthetic foot.			Outcome	A portable prosthetic foot is realised, its structure is designed by topology optimisation and its structure is verified by finite element analysis, it is lightweight and the cost of the foot is significantly reduced when carrying out batch or mass production processes.				

III. RESULTS

A total of 8 articles were included following the screening process, with these studies spanning the period from 2009 to 2021. These studies focus on different types of prostheses, namely: prosthetic hand (2 studies, [10], [11]), prosthetic finger (2 studies, [12], [13]); prosthetic foot (2 studies, [14], [15]); and transtibial prosthetic socket (2 studies, [16], [17]). With regard to the purpose of the use of the Generative Design Tools, 3 studies looked at the use of parametric design to help prosthetic technicians modify the design of prostheses without the need for specialist 3D modelling training [10,13,16], 2 studies use Topology Optimization to reduce the weight of 3D printed prostheses [14,15], 3 studies use FEM simulation to validate the design of the prosthesis [11,12,17]. All 8 studies used Fused Deposition Modelling (FDM) print technology, varying in the print materials they used, 3 studies used Polylactic Acid (PLA) [13–15], 2 studies used Thermoplastic Polyurethane (TPU) [11,12], 1 study used Acrylonitrile Butadiene Styrene (ABS) [10] and 2 studies did not state the print material they used but did mention the use of FDM print technology [16,17]. Three studies have post-processed the printed parts to strengthen them, two of these studies wrapped the printed prostheses with a layer of resin [16,17] and the third added metallic elements to the printed parts [11]. There is a lack of consistency in the CAD modelling software and simulation tools used in these studies, all 6 studies used CAD software to create the initial digital models, with the exception of 2 studies [16,17] which used laser scanning equipment for initial modelling.

IV. DISCUSSION

This review explores the current state of knowledge in the field of generative design in 3D printed limb prostheses. 3D printed limb prosthesis in itself is a relatively emerging field, the state of knowledge is limited, existing research has only a small sample, and many of its approaches lack systematic justification and standardisation [3], which is further compounded by the limited practice of 3D printed prostheses using generative design methods. Of the 125 studies that used 3D printing technology to manufacture limb prostheses, only 8 studies utilised the generative design approach. Yet the application of Generative Design in other areas of 3D printing, such as aircraft [18] and motorsport [19] manufacturing, could also be applied to the field of prosthetics and ameliorate the problematic issues in this area. Generative design uses algorithms to generate a set of design options for evaluation, and parametric design uses parameters and rules to create design solutions that can be easily modified. Generative design uses algorithms to produce design results, and if these algorithms rely on a set of parameters, it can also be considered a parametric design [20]. Five of the studies analysed in this review used this approach to improve the physical properties of prostheses [11,12,14,15,17], and three used it to optimise the prosthetic design process [10,13,16].

A. Generative Design for better low-cost 3DP Limb Prosthetics (LP)

The two factors that are of most concern to amputees are the weight and the cost of the prosthesis [6]. While 3D printing

reduces the cost of manufacturing prostheses, it also loses the advantages of weight and robustness compared to traditional casting and moulding methods [3]. Researchers have therefore tried to address these issues in various ways, most commonly by using Topology Optimisation to refine the morphology of prosthetic printed parts [14], in order to reduce weight [15], and material consumption. By post-processing the printed parts with reinforcing materials [11,17], the 3D printed prostheses can also be made to an acceptable level of robustness. 3D printed soft monolithic robotic hands are also easier to design through FEM analysis, and this type of prosthetic hand is more effective and does not rely on as many sensors when manipulating fragile objects compared to traditional rigid prosthetic hands [12].

B. Generative Design for the easily customisable 3DP Limb Prosthetics (LP)

Generative design solves one of the major problems faced by low-cost, adaptive prostheses, namely that these open-source designs can often only be scaled to fit the size of the user as a whole but cannot be fine-tuned precisely for each individual user, modifications to prosthetic designs often require specialist 3D modelling training, which is not accessible in many clinics [16,21]. Using a parametric design approach, the prosthetic model can be adjusted by altering the dimensions of each component to fine-tune the overall design [13], and this can be done using a graphical interface [16] or editing spreadsheets [10]. This feature could be useful when performing remote fitting as it would allow the user to make their own adjustments to the prosthesis design [22]. Tinker Space [23] or projects from the open-source community like the e-NABLE project [24] will also be able to help more prosthetics users since the design barriers are reduced.

V. CONCLUSION

Generative Design is a way to effectively reduce the weight and increase the robustness of 3D printed prostheses [11]. It allows prostheses to be manufactured with inexpensive 3D printing equipment [11], and by combining parametric design, it enables prosthetic technicians, without professional modelling skills, to create deeply customised 3D printed prosthetic designs for amputees and effectively improve the user experience [10].

However, the state of knowledge in this area is still very limited and more research is needed to fully understand the impact of this technology on patients and the healthcare system. Furthermore, the reliability of this approach requires systematic analysis with larger samples to validate.

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